Data Lakehouse Project Documentation

Medallion Architecture Implementation Report

1. Architectural Overview

This project implements a three-tier **Medallion Architecture** (Bronze, Silver, and Gold Layers) to clean, standardize, and analyze a transactional sales database using PySpark.

Layer	Path	Purpose
Bronze	data_lake/bronze/	Raw, untouched ingestion storage.
Silver	data_lake/silver/	Validated, cleaned, and standardized schema.
Gold	data_lake/gold/	Aggregated business KPI summaries and reporting.

2. In-Depth Pipeline Execution Stages

Phase A: Bronze Ingestion Layer

Establishes the initial landing zone. The raw dataset sales (3).csv was imported directly with no schema modifications to preserve data history.

Phase B: Silver Cleaning Layer

Transforms raw data into an analytical-ready state by addressing four key categories of anomalies:
• **Duplicates**: Removed matching row duplicates (reducing from 1000 to 990 total rows).
• **Whitespaces**: Dynamically trimmed all string columns.
• **Outliers**: Coerced negative prices/quantities to 0 and limited discounts to 0%-100%.
• **Dates**: Standardized disparate date strings to clean ISO-8601 formatting.

Phase C: Gold Analytics Layer

Calculates aggregated business metrics. Net Revenue was calculated dynamically at line-item level using: Net Revenue = Quantity * Unit Price * (1 - Discount/100), then aggregated by product category.

3. Key Findings & Category Performance Summary

Category	Units Sold	Total Revenue ($)	Avg Discount (%)	Transactions
Furniture	704	8,639,378.50	21.13%	129
Electronics	549	7,270,942.70	17.70%	103
Home Appliances	596	2,761,348.40	23.62%	113
Sports	737	1,651,746.80	19.80%	136
Clothing	557	1,341,868.60	18.51%	107